

Quantifying Uncertainty in Latency-Aware Consensus Protocols

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Abstract

Work in computing research has long relied on assumptions we test directly here. The measurements reported here were collected over twelve independent trials under controlled conditions. Calibration drift over the trial period remained below the threshold set out in the protocol. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

1. Introduction

Calibration drift over the trial period remained below the threshold set out in the protocol. Sensitivity to the smoothing window proved modest across the range we examined. Replicate agreement was high, suggesting the protocol is robust to small handling differences. We release the processing scripts and intermediate artefacts to support independent verification. The measurements reported here were collected over twelve independent trials under controlled conditions. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. Residual error grows approximately linearly with the number of coupled parameters. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

The reduced model retains most of the explanatory power at a fraction of the computational cost. Calibration drift over the trial period remained below the threshold set out in the protocol. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Residual error grows approximately linearly with the number of coupled parameters. Neither of the alternative formulations improved on the simple additive model in our setting. Sensitivity to the smoothing window proved modest across the range we examined. We release the processing scripts and intermediate artefacts to support independent verification.

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computational cost.

2. Related Work

The measurements reported here were collected over twelve independent trials under controlled conditions. Calibration drift over the trial period remained below the threshold set out in the protocol. Neither of the alternative formulations improved on the simple additive model in our setting. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

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3. Materials and Methods

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4. Experimental Setup

Residual error grows approximately linearly with the number of coupled parameters. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Neither of the alternative formulations improved on the simple additive model in our setting. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Taken together, these findings argue for treating the two regimes separately in future analyses. The measurements reported here were collected over twelve independent trials under controlled conditions.

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5. Results

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